COP2800C Module 13 Graded Programming Assignment – Database Access

In this assignment, we will use JDBC to create a database and table to store our Palmer Penguins data. We will read the data from the DB and display it, then drop the table and database

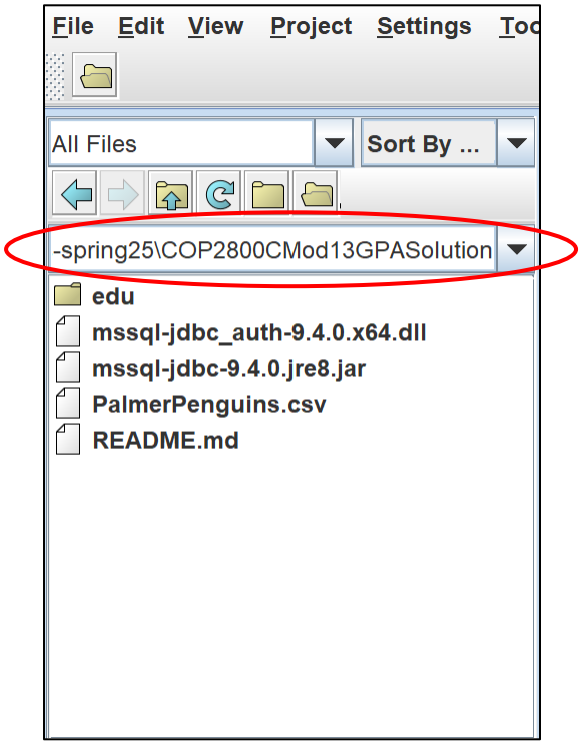
I recommend that you use the Horizon system for this assignment unless you are comfortable working with Microsoft SQL Server. All necessary software is already in place on the Horizon system.

Follow these steps to complete this assignment.

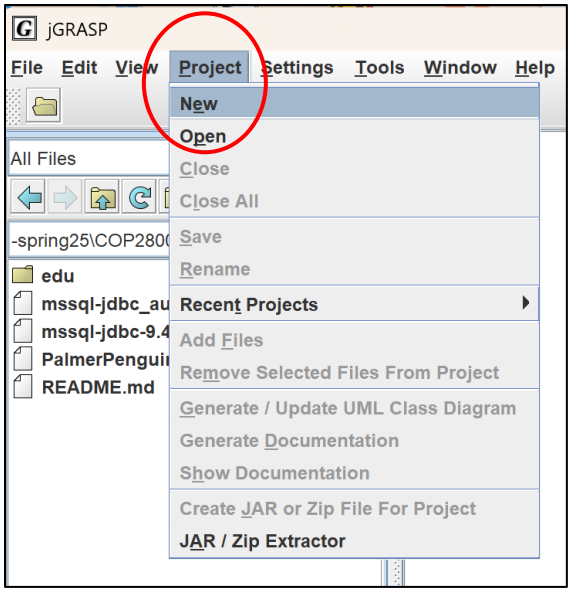
1. Accept the GitHub Classroom Invite

- Clone your private fork. The repository contains the CSV data and the required starter files.

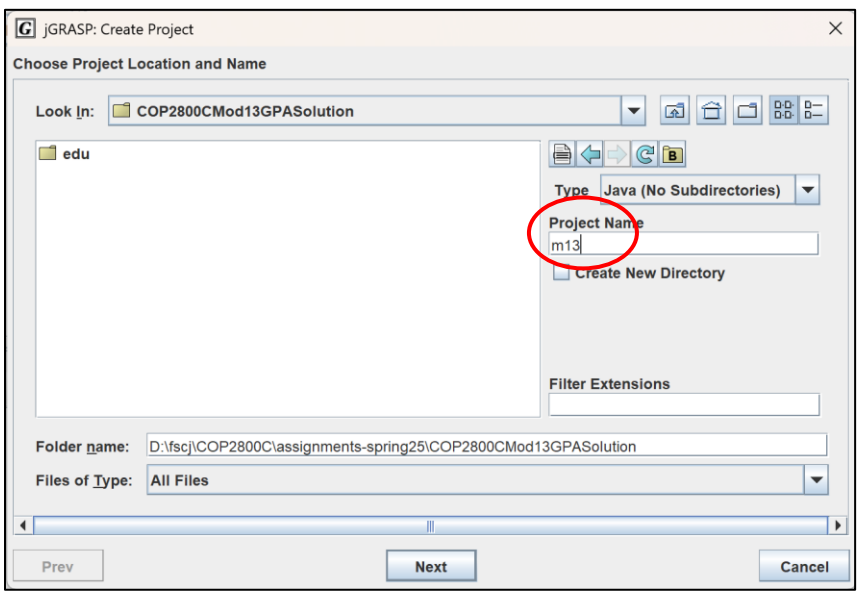
Navigate to the top level of your cloned repo to access the starter files:



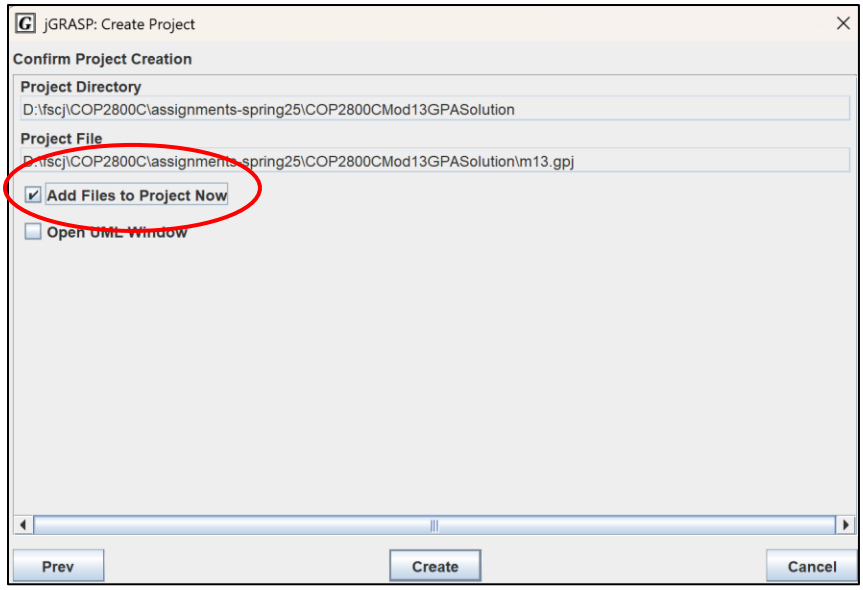
Create a new project:



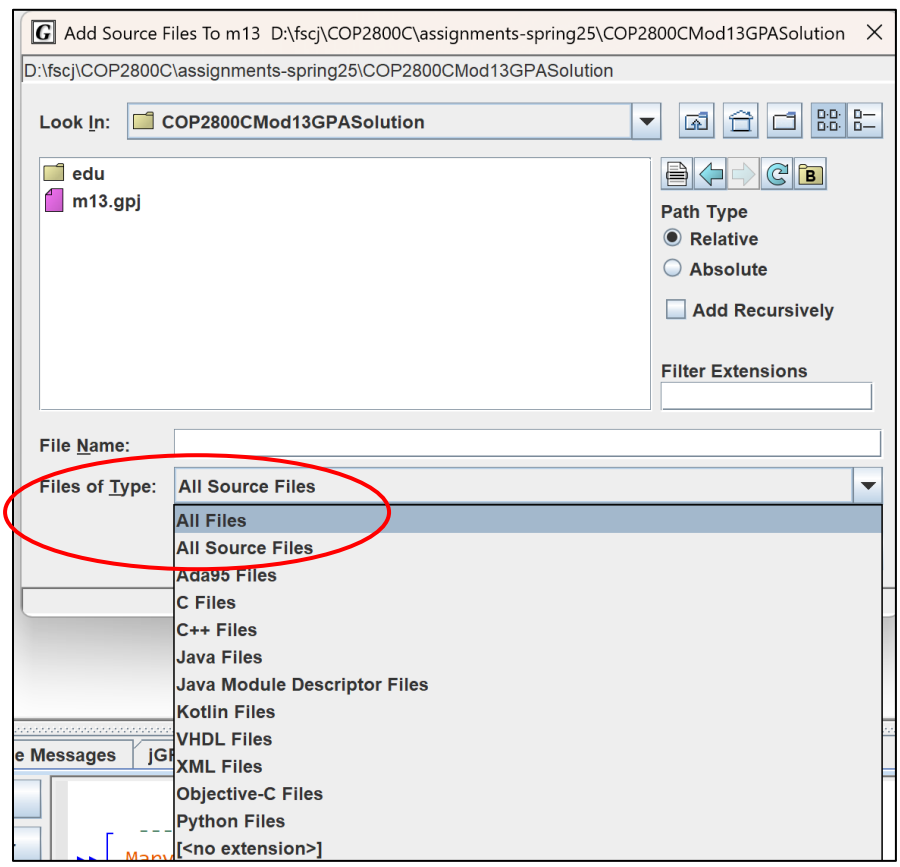
Name it m13 and click Next:



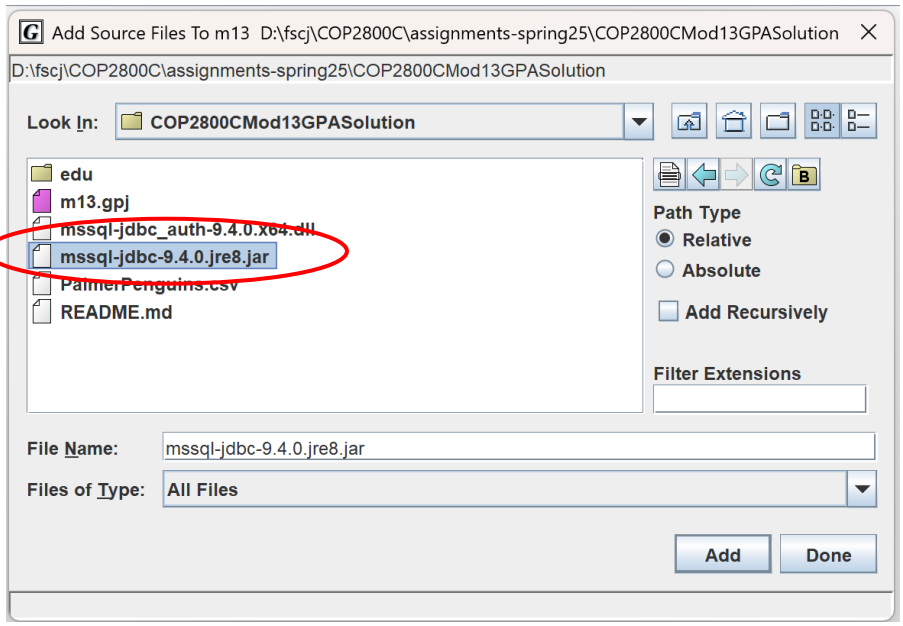
Check **Add Files to Project Now** and click Create:



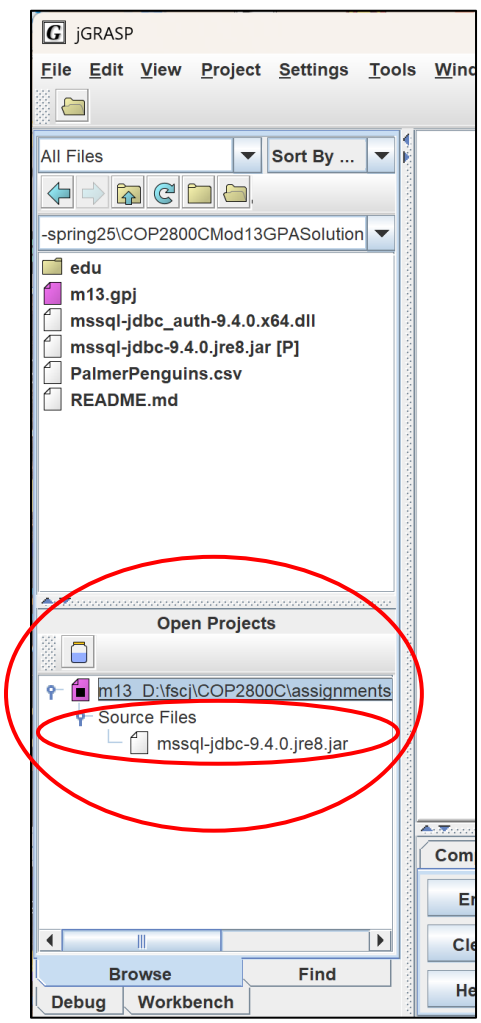
Select **All Files** from the **Files of Type** drop-down in the Add Source Files dialog:



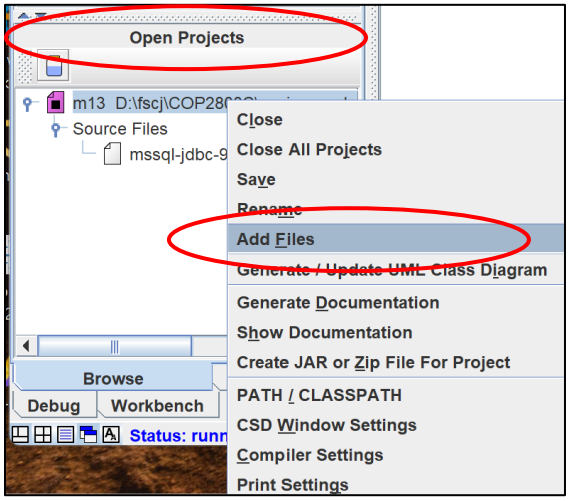
Select the mssql-jdbc-9.4.0.jre8.jar file and click Add (do not select any other files yet):



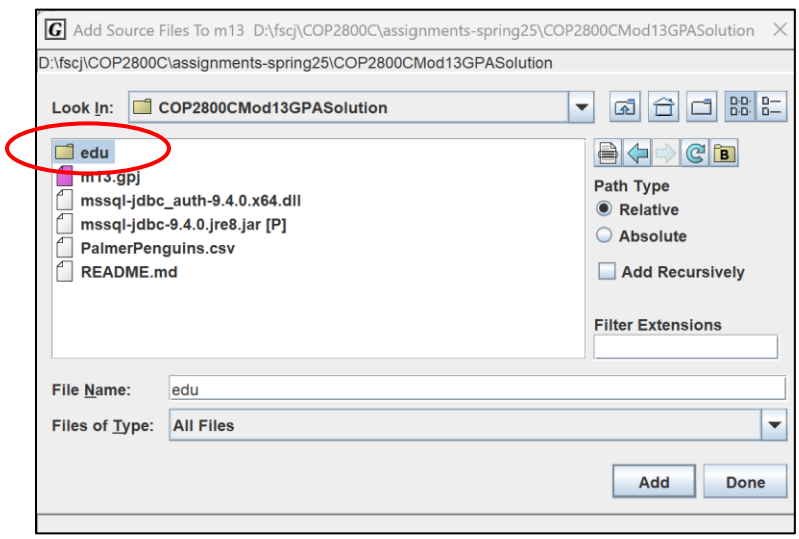
In the jGrasp **Open Projects** pane, verify the jar file has been added:



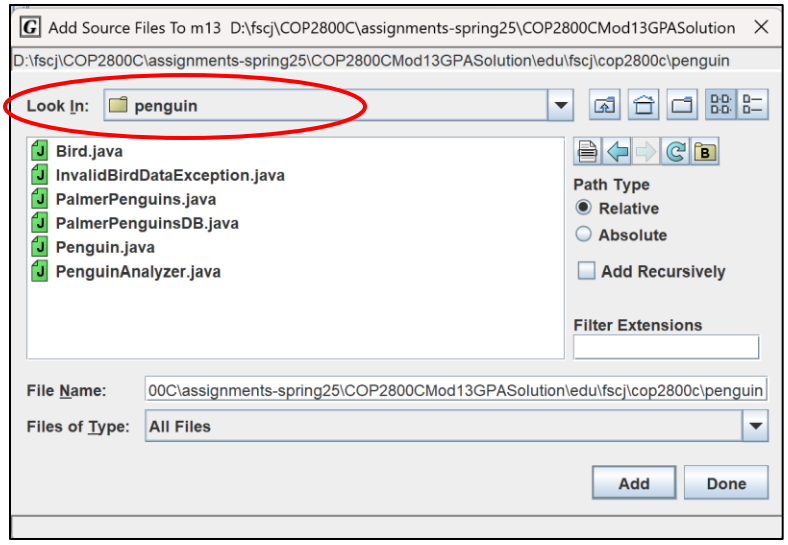
Right-click on the project name in the **Open Projects** pane and select **Add Files**:



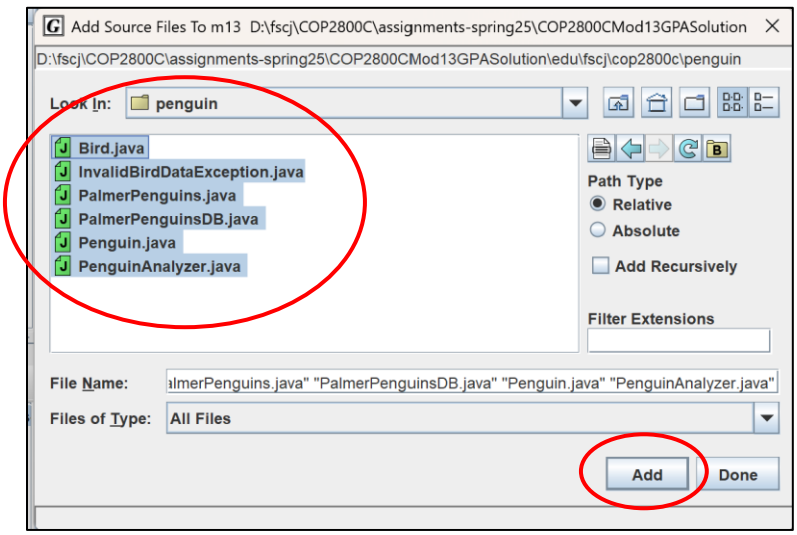
Double click on **edu** to navigate down into that folder:



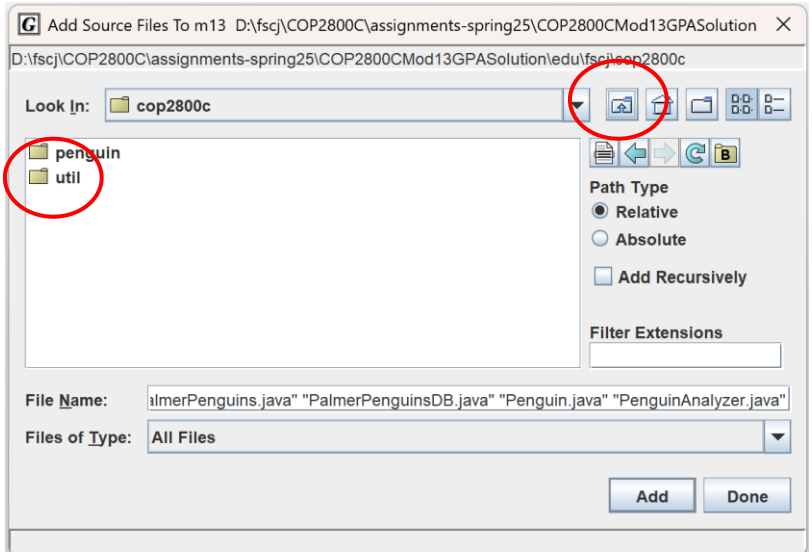
Keep double-clicking on the subfolders until you have navigated down to the **penguin** folder (sub-package):



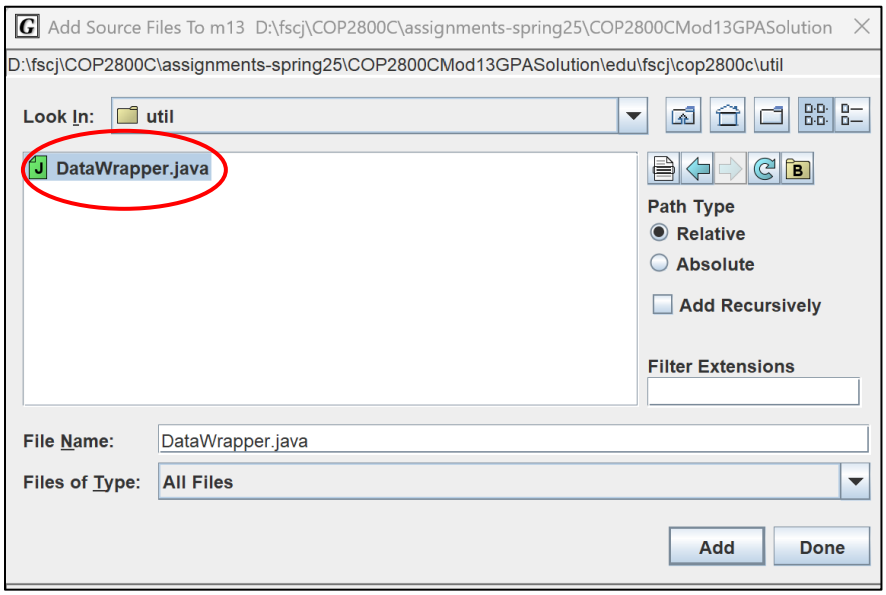
Press Ctrl-A to select all the files in this folder, then click **Add**:



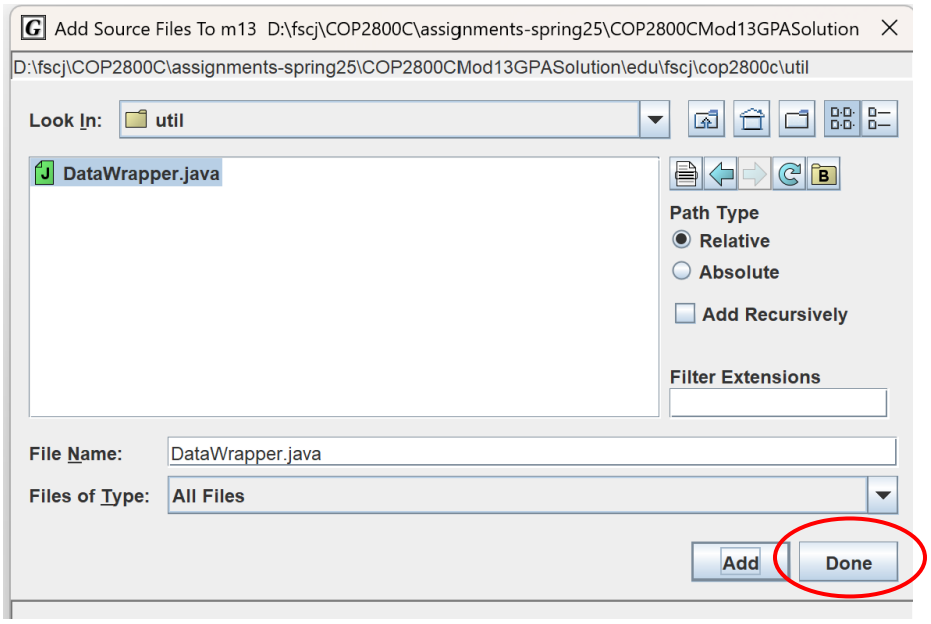
Use the"Up Folder" button to navigate up one folder level, then double click on the **util** folder to navigate down into it



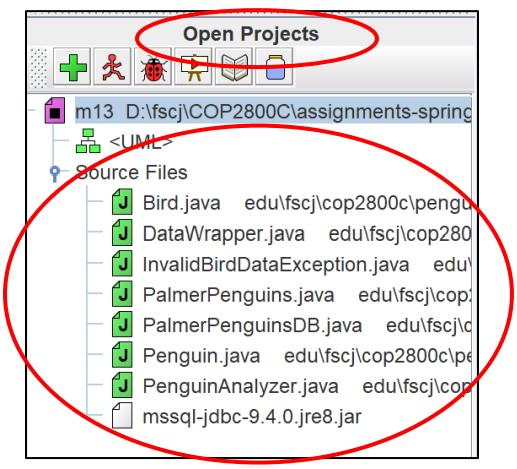
Select the DataWrapper.java file and click **Add**:



Click **Done**:



In your **Open Projects** pane verify you have correctly added a total of eight files as shown here:



3. You can now select files for editing directly from the Open Projects pane now instead of navigating down into the sub-packages. Select and modify the PalmerPenguinsDB.java file by filling in the missing code as follows:

Find the comments marked

```
// *** add your code here
```

and replace them with your working code (there are 2 of these in the file; one for the table creation, and one for the data query and display).

For the table creation:

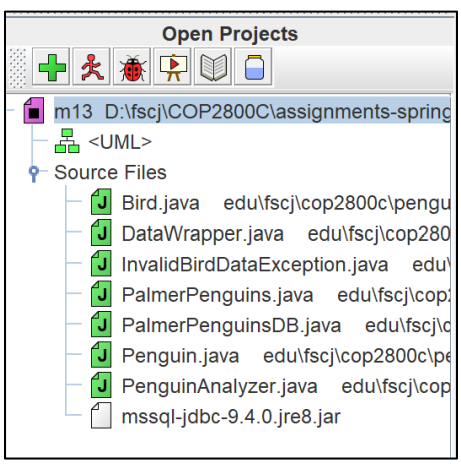
```
String createTable = "CREATE TABLE Penguin " +
        "(SAMPLENUM smallint PRIMARY KEY NOT NULL," +
        "CULMENLEN float NOT NULL," +
```

```
            "CULMENDEPTH float NOT NULL," +
            "BODYMASS smallint NOT NULL," +
            "SEX char(1) NOT NULL," +
            "SPECIES varchar(20) NOT NULL," +
            "FLIPPERLEN float NOT NULL)";
```

For the data query and display:

```
while (rs.next()) {
    System.out.println(
        rs.getInt("SAMPLENUM") + "," +
        rs.getDouble("CULMENLEN") + "," +
        rs.getDouble("CULMENDEPTH") + "," +
        rs.getInt("BODYMASS") + "," +
        rs.getString("SEX") + "," +
        rs.getString("SPECIES") + "," +
        rs.getDouble("FLIPPERLEN")
    );
}
```

4. Build and execute the project in jGrasp. You can also use the Open Project pane for this, the tools are the same as those on the jGrasp top menu bar; click on the green plus sign to build the project and the red running man to execute (or to build and execute in one step).



5. Take **two** screenshots of the output pane in jGrasp, one showing the DB and table being created and the first part of the data being read, and then the end of the output showing the last part of the data (see Expected Output below). Submit your project and screenshots to the repository using **Git**.

Expected Output:

```
DB created
Table created
Reading from DB
1,39.1,18.7,3750,M,ADELIE,181.0
```

```
… <<< 340 lines not shown
342,49.9,16.1,5400,M,GENTOO,213.0
Penguin table dropped
could not drop DB, in use        <<< ignore this if you see it
```